Shren Patel

Senior Backend Engineer — AI Agent Runtime

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SUMMARY

10+ years engineering with platform + infra depth. Built CentrOS runtime — durable execution, session/conversation state, checkpointing, crash recovery, autoscaling, versioned rollouts, LLM streaming — 0 silent run loss across 6 months; 3,980+ Vitest tests on main. Strong Python + Node/TypeScript; comfortable owning on-call for real-time AI.

EXPERIENCE

Founder CentrOS — AI Agent Orchestration Platform *Feb 2026 – Present*

Remote · Canada • TypeScript · Hono · PostgreSQL · Drizzle · Vitest · LLM tool-calling

- Built CentrOS runtime: durable workflow execution, checkpointing, crash recovery, retries, versioned rollouts, LLM streaming — Temporal-style semantics for live agent workflows; 0 silent run loss over 6 months.
- Designed runtime contracts beneath agent behavior: explicit state machines, durable storage, event sourcing, idempotent retries, observability across 3 span types (streaming/tool/state) — Kubernetes-native autoscaling.
- Built eval harness + LangFuse trace pipeline: golden datasets, task-level scorers, regression suites, prompt versioning, failure taxonomy — caught 22+ silent regressions before release.
- Designed HITL interrupt model: confidence-gated handoff, approval checkpoints before irreversible actions, reversible operations, audit trails, escalation policy — 100% of high-risk tool calls human-reviewed.

Vertical automation & ops dogfood Conscious Creations *Sep 2022 – Present*

Remote · Canada • Python · FastAPI · LLM tool-calling · workflow automation

- Designed AI-to-human handoff: shared state contract, full conversation history, durable IDs, operator-readable summaries — 0 lost context across 100% of escalation events in live ops.

Senior Frontend Developer Faire *Jun 2021 – Aug 2022*

Remote · Canada • React · TypeScript · MobX · Jest · Storybook · A/B testing

- Drove site-wide perf gains via A/B testing + staged rollouts; same guardrail discipline now applied to agent versioning, shadow traffic, and canary releases at CentrOS — 5+ wins shipped.

Software Developer Minted LLC *Aug 2014 – Dec 2014*

San Francisco • Python · image processing · color space conversion · AWS

- Built conversion-loss trackers + shipped 3+ hotfixes during Cyber Monday peak (10x+ traffic spike) — early production-incident response discipline.

SKILLS

Runtime & Distributed Systems:

- Stateful runtime · session state · checkpointing · crash recovery · replay
- LLM streaming · partial output handling · timeouts · cancellation · backpressure
- Distributed systems · Kubernetes · autoscaling · versioned rollouts · canary
- Observability: trace spans · latency histograms · failure taxonomy · incident response
- Workflow engines: Temporal · LangGraph state graphs

Languages:

- Python · Go (familiar) · TypeScript · Node.js

Infra:

- GCP · AWS · Kubernetes · Docker · serverless · IaC · CI/CD

EDUCATION

University of Waterloo Bachelor of Software Engineering (BSE) — Computer Software Engineering *2012 – 2018*

Waterloo, ON